

MITRE ATT&CK; Mapping

Project: Brute Force & Account Compromise Investigation Lab

This document maps the simulated attack activities performed during the Windows Attack Detection Lab to the MITRE ATT&CK; framework. The mapping demonstrates how endpoint telemetry collected through Windows Security Logs, Sysmon, Splunk Enterprise, Microsoft Defender for Endpoint, and Microsoft Sentinel corresponds to adversary tactics and techniques.

Tactic	Technique	ATT&CK ID	Evidence Collected	Detection Source
Initial Access	Brute Force	T1110	Repeated Event ID 4625 failed logons	Windows Security, Splunk, Sentinel
Initial Access	Valid Accounts	T1078	Successful Event ID 4624 logon	Windows Security, Splunk
Execution	PowerShell	T1059.001	powershell.exe execution	Sysmon, Defender, Splunk
Discovery	Process Discovery	T1057	tasklist	Sysmon, Splunk
Discovery	System Information Discovery	T1082	hostname	PowerShell
Discovery	Account Discovery	T1033	query user	PowerShell
Discovery	Permission Groups Discovery	T1069	whoami /groups	PowerShell
Discovery	Local Group Discovery	T1069.001	net localgroup administrators	PowerShell
Discovery	Network Configuration Discovery	T1016	ipconfig /all	PowerShell
Discovery	Network Connections Discovery	T1049	netstat -ano	PowerShell

Analysis

The investigation demonstrates a complete authentication attack chain beginning with brute-force attempts, progressing to successful use of valid credentials, followed by administrative PowerShell execution and host discovery. Correlation across Splunk, Microsoft Defender, and Microsoft Sentinel confirmed consistent visibility of the simulated activity. The observed behaviour aligns with common post-compromise techniques documented by the MITRE ATT&CK; framework and illustrates effective SOC investigation and threat hunting workflows.

Recommendations

Implement multi-factor authentication, enforce account lockout policies, monitor excessive Event ID 4625 activity, alert on PowerShell execution, review Event IDs 4672 and 4688, and continuously map detections to MITRE ATT&CK; to improve coverage.